Why Students Order Out

Analysing Mess Food Habits and Outside Food Ordering Habits among Hostel Students

IDHV 2026M Project Report | Introduction to Data Handling and Visualization

Group number	7
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Data collected	Online survey, 12 Oct 2026 to 16 Oct 2026; 59 respondents

Abstract

We surveyed 59 hostel students about two things: how they experience the mess, and how they order food from outside. Using descriptive statistics and visual exploratory analysis, we found that dinner is the least-liked mess meal (75% dissatisfied) and also the meal that 92% of students order outside food for. A typical student orders 2 times a week and spends about Rs 300, roughly Rs 150 per order, and weekly spending rises steadily with the number of orders. Ordering is mostly social (67% of orders are with friends), and it is rare on Wednesdays and Sundays, the days the mess serves non-veg meals. Skipping mess meals, disliking the menu and low mess ratings tend to go together with more ordering and higher spending, although these links are weaker than the link between the number of orders and spending.

1. Introduction

1.1 Motivation

Hostel students already pay for the mess, yet many of them regularly pay again for food from outside. For students this is a recurring expense, and for mess management it is a signal that something about the mess is not working. Understanding *when*, *how often*, *with whom* and *why* students order outside food is a first step to understanding where the mess falls short and how much money students spend as a result.

1.2 Research question

Main question: How do students' mess food habits and outside food ordering habits look on their own, and how are the two related?

- **Mess habits alone:** How satisfied are students with each mess meal, with taste and quality? How many meals do they skip, and how often do they dislike the menu?
- **Outside ordering habits alone:** On which days, how often and with whom do students order? How much do they spend? What reasons do they give?
- **Relationships:** Is the least-liked meal the one students replace? Do more orders mean more spending? Do skipping meals, a disliked menu or low satisfaction go together with more ordering?

1.3 Objectives

- Design a short questionnaire and collect responses from more than 50 students (quantitative and qualitative variables).
- Clean and organise the raw data: encode, handle outliers, derive new variables and normalise where needed.
- Describe each habit with simple summary statistics and charts, and uncover relationships with visual exploratory data analysis.

2. Methodology

2.1 Research design and sampling

We ran a cross-sectional online survey (Google Forms) among hostel students of IIITS who eat in the mess. The form was shared in student groups and collected 59 responses between 12 Oct 2026 and 16 Oct 2026. The sampling method was **convenience sampling**, in which responses are obtained from members of the target population who are readily accessible. It was adopted because no sampling frame listing hostel residents was available, and because the study had to be completed within the duration of the course. The method does not yield a probability sample, and the achieved sample is concentrated in one cohort: 50 of 59 respondents (85%) are third-year students. The findings therefore describe the respondents and should not be generalised to the hostel population as a whole. Ordering and mess habits were asked about a *typical week* so that respondents could answer from routine rather than recall. No personally identifying information was collected: participation was voluntary and all responses are anonymous.

Context used in the analysis: the mess serves non-veg meals (for example chicken curry and biryani) on Wednesdays and Sundays. This is the group's knowledge of the weekly menu; the survey did not ask about it directly.

2.2 Questionnaire and variables

The questionnaire has three parts: (i) outside ordering behaviour, (ii) mess satisfaction and habits, and (iii) year of study. It mixes single-choice, multiple-choice, 5-point rating and numeric questions. The full list of questions is in the Appendix.

Variable group	Variables	Type
Outside ordering (quantitative)	Weekly spend (Rs); orders per week; orders alone; orders with friends	Numeric (ratio / count)
Outside ordering (qualitative)	Usual ordering company; reasons (multi-select); meal ordered for; days ordered (multi-select); ordered outside mess hours; ordered after skipping a meal	Nominal / binary
Mess satisfaction	Taste (1-5); quality and consistency (1-5); satisfaction with breakfast, lunch, snacks, dinner (5-point scale)	Ordinal
Mess habits	Meals skipped per week; days menu disliked (of 7); how often the menu influences ordering (4 levels); favourite meal	Count / ordinal / nominal
Background	Year of study	Nominal

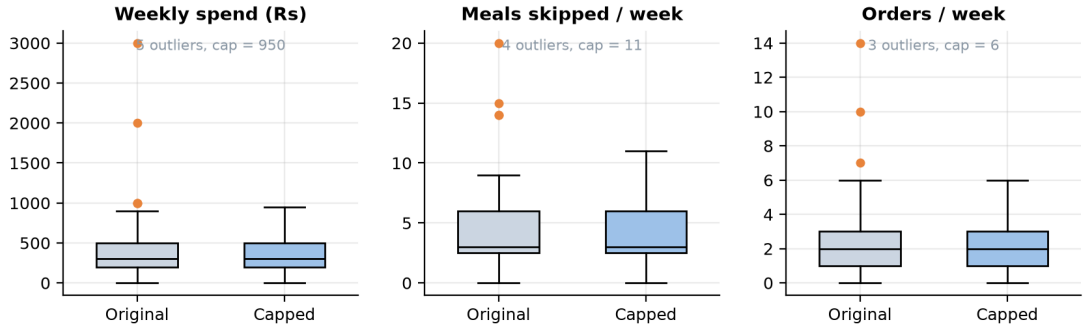
The survey therefore has 22 analysis variables and 59 respondents, which satisfies the project requirement of more than 50 data points and more than 3 variables with both quantitative and qualitative types.

3. Data Preprocessing

Step	What was done	Why
1. Inspection	59 responses and 23 columns checked: 4 blanks and 0 duplicate responses found.	Establishes what has to be repaired before analysis.
2. Tidying	Columns matched on question text and renamed to short names; timestamp dropped from the cleaned file; text and numeric answers trimmed (some were entered as "2 " with a trailing space).	Easier analysis, and the script still loads a re-export whose columns are in a different order.
3. Encoding	Satisfaction answers coded 1 (very dissatisfied) to 5 (very satisfied); menu influence coded 1-4; Yes/No coded 1/0.	Needed for averages and correlations.
4. Missing values	4 blank entries were found, all in Question 18 (days on which the menu is disliked). No other question had a missing value. Each blank was replaced by the median of that column (3 days) and marked in an indicator column, disliked_days_imputed. No response was blank in more than half of its fields, so no response was removed (0 deleted).	Median imputation keeps the respondent's remaining answers available to the analysis, which deletion would discard. The median is preferred to the mean because the variable is a bounded day count (0-7) and mildly skewed. The indicator column allows the analysis to be repeated with the affected responses excluded.
5. Multi-select split	Question 4 (days on which outside food is ordered) was split into 7 binary columns, one per day of the week. Question 2 (reasons for ordering) was split into 9 binary columns, one per listed reason.	Both questions allow more than one answer, so a single cell may hold several options separated by semicolons. Binary indicator columns make each option countable and allow combinations to be compared.
6. Consistency rules	**Rule 1 (weekly order total).** Questions 15, 16 and 17 ask separately for the weekly total, the orders placed alone, and the orders placed with friends. The three answers agree in 46 of 59 responses. In the remaining 13, the stated total was replaced by (orders alone + orders with friends). **Rule 2 (orders after skipping a meal).** Question 8 asks how many of those weekly orders followed a skipped mess meal. These are part of the weekly total and so cannot exceed it. In 3 responses the count was larger than the total, and it was reduced to that respondent's own total. The values as answered are retained in the columns orders_as_stated and orders_after_skip_as_stated.	An outside order placed by a hostel student is either placed alone or with friends, so the two parts together must account for the weekly total. Where the three answers disagree, the two part-counts are the better supported figure, since each is given separately and the total is a summary the respondent has to work out. Rule 2 applies the same part-and-whole logic.
7. Derived variables	Spend per order; share of orders with friends; overall mess index (mean of taste, quality and four meal ratings); number of order days; ordered on a non-veg day; order, skip and mess-satisfaction groups.	Simple, interpretable measures. The six ratings are consistent with each other (Cronbach's alpha = 0.76).
8. Outliers	IQR rule (above Q3 + 1.5 x IQR). Weekly spend: 5 values above Rs 950 (1000, 1000, 1000, 2000, 3000). Meals skipped: 4 above 11.2 (14, 14, 15, 20). Orders: 3 above 6 (7, 10, 14). Values were capped at the fence in new columns ending in _capped; original columns were kept.	With only 59 responses, deleting students would lose information. The Rs 3000 answer could be a typing error or a genuine heavy spender, so we limit its influence rather than drop it.
9. Normalisation	Min-max scaling (0 to 1) of capped spend, capped orders, capped skips, disliked days and the mess index.	Puts variables with different units on one scale so they can be compared in one chart (Figure 17).

Descriptive statistics in Section 4 use the **original** values; the scatter plots and the correlation heatmap use the **capped** values so that a few extreme answers do not dominate.

Figure 1. Outlier treatment (IQR rule): before and after capping



4. Results and Analysis

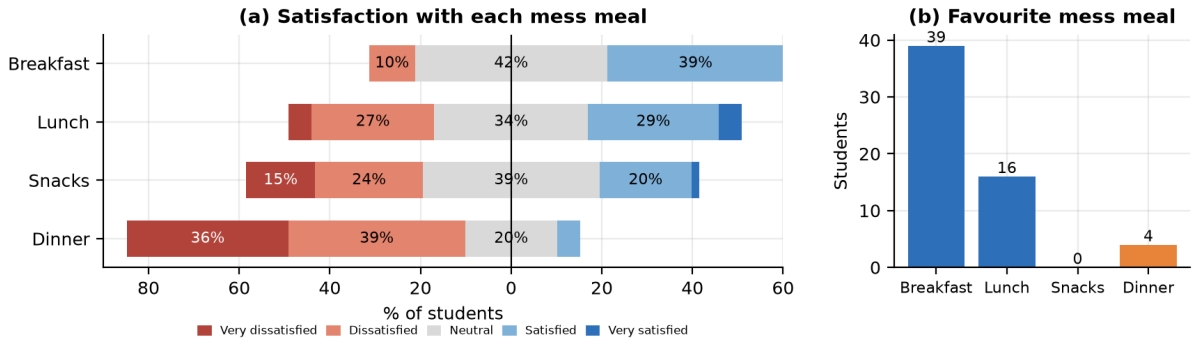
4.1 Summary statistics

Variable	Mean	Median	Std dev	Min	Max
Weekly spend (Rs)	437.3	300.0	463.2	0.0	3000.0
Orders per week	2.7	2.0	2.3	0.0	14.0
Orders alone	0.9	1.0	1.4	0.0	7.0
Orders with friends	1.8	2.0	1.4	0.0	7.0
Spend per order (Rs)	167.7	150.0	89.4	10.0	500.0
Orders after skipping a meal	0.8	0.0	1.2	0.0	5.0
Meals skipped per week	4.6	3.0	3.8	0.0	20.0
Days menu disliked (of 7)	3.5	3.0	1.3	1.0	7.0
Taste rating (1-5)	2.19	2.00	0.82	1.00	4.00
Quality rating (1-5)	2.41	2.00	0.81	1.00	4.00
Overall mess index (1-5)	2.62	2.50	0.60	1.33	4.50

Weekly spend is strongly right-skewed: the mean (Rs 437) is well above the median (Rs 300) because a few heavy spenders pull it up, so we quote the **median** as the typical value. A typical student orders 2 times a week, skips 3 mess meals and dislikes the menu on 3 of 7 days. Taste (mean 2.19) and quality (mean 2.41) are both below the mid-point of 3.

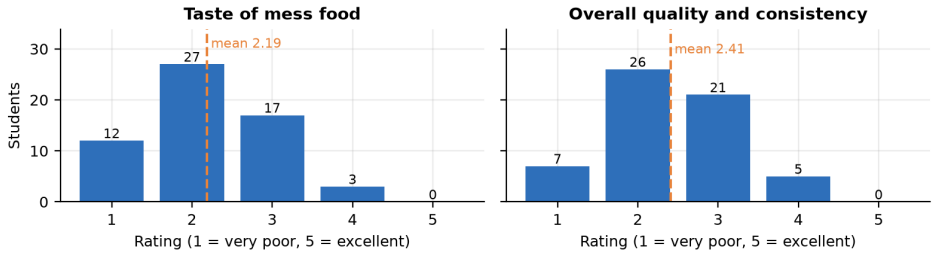
4.2 Mess habits on their own

Figure 2. Mess meals: satisfaction and preference



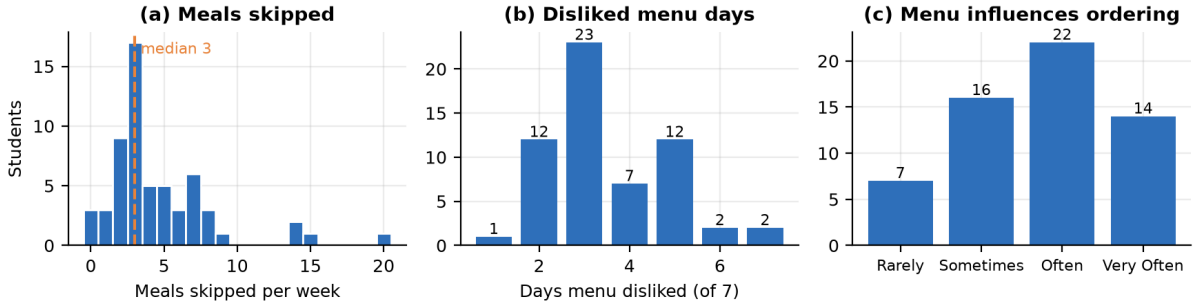
Observation. Mean satisfaction falls across the day: 3.46 for breakfast, 3.02 for lunch, 2.69 for snacks and 1.95 for dinner. 6 (10%) students are dissatisfied with breakfast, compared with 44 (75%) students for dinner; only 3 students are satisfied with dinner. Breakfast is the preferred meal of 39 (66%) students, lunch of 16, dinner of 4, and snacks of none.

Figure 3. Taste and quality ratings



Observation. 39 (66%) students rate the taste of mess food 1 or 2 on the 5-point scale, and 33 (56%) students rate overall quality and consistency 1 or 2. No student awarded the highest rating of 5 to either. The two ratings move together (rank correlation 0.54).

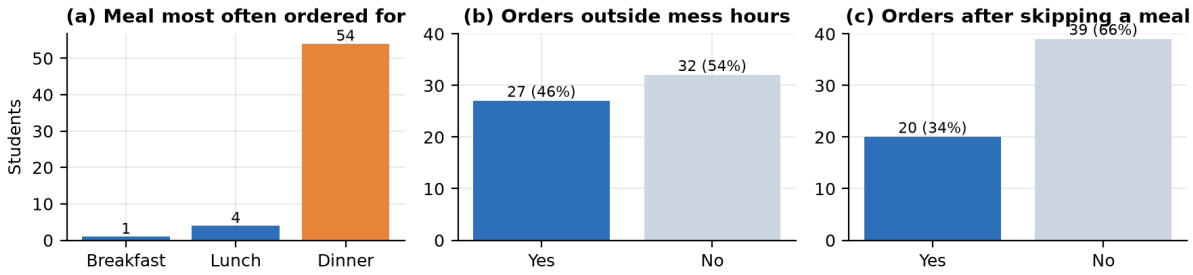
Figure 4. Mess habits



Observation. Meal skipping is common: the median is 3 meals a week, 17 (29%) students skip six or more, and 3 students skip none. The menu is disliked on 3.5 days a week on average, and 16 (27%) students dislike it on five or more days. 36 (61%) students report that the daily menu influences their decision to order outside food Often or Very Often.

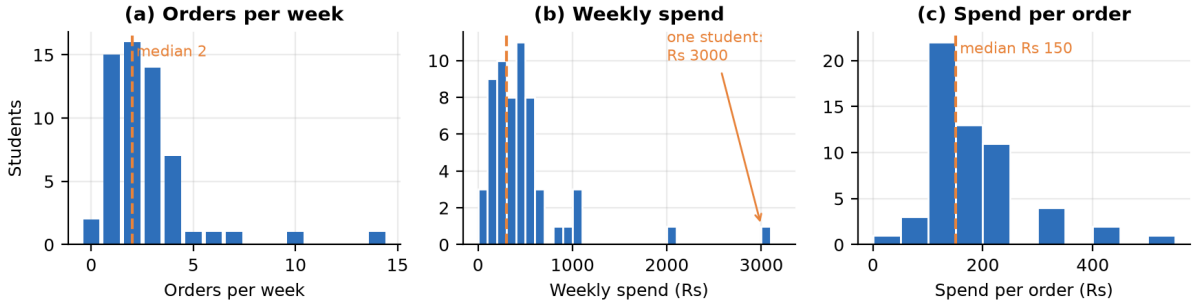
4.3 Outside ordering habits on their own

Figure 5. When and why-context of ordering



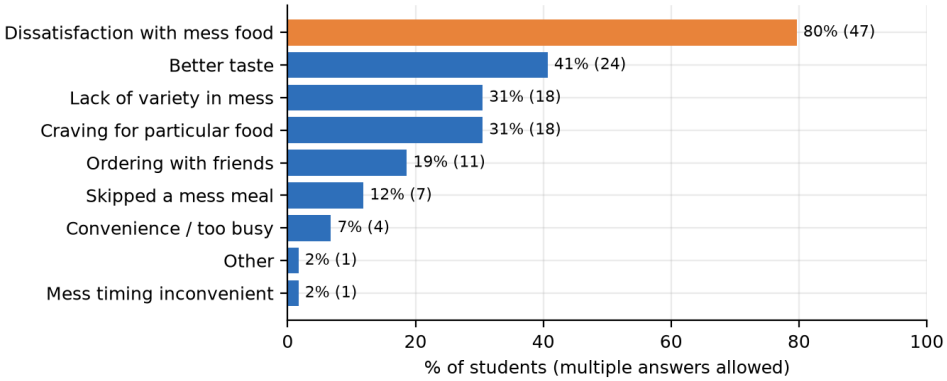
Observation. Outside ordering is concentrated at dinner: 54 (92%) students order mainly for dinner, compared with 4 students for lunch and 1 for breakfast. 27 (46%) students usually order outside mess timings, and 20 (34%) students order because they skipped a mess meal.

Figure 6. Ordering frequency and spending



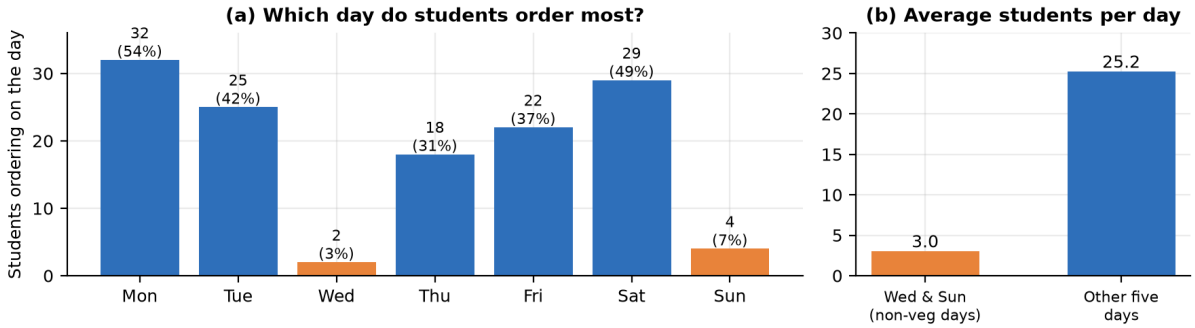
Observation. Weekly order counts are low for most students: 2 students order none, 15 order once, 16 twice, 14 three times and 12 four or more times. The typical student spends Rs 300 a week (about Rs 1200 over four weeks), or Rs 150 per order. 18 (31%) students spend Rs 500 or more, and the six highest spenders account for 34% of the Rs 25800 spent by the whole group in one week.

Figure 7. Reasons for ordering outside food



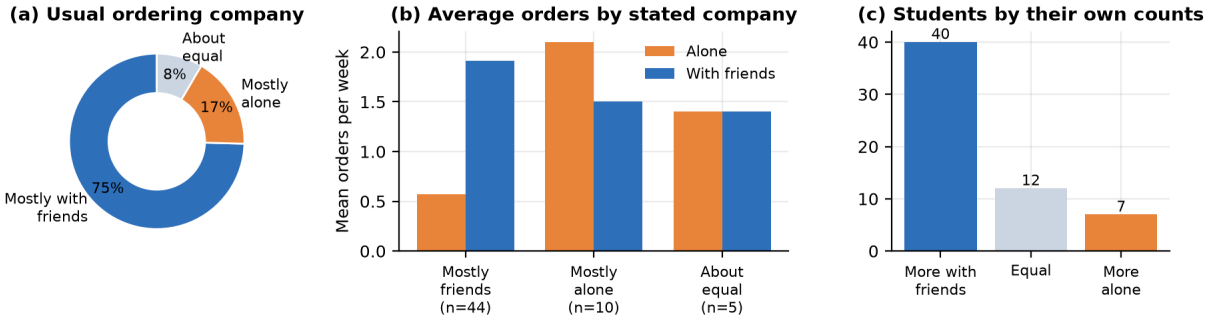
Observation. Dissatisfaction with mess food is the most frequently cited reason, given by 47 (80%) students, followed by better taste (41%), craving for a particular food (31%) and lack of variety (31%). Practical reasons are rare: convenience is cited by 4 students and inconvenient mess timing by 1.

Figure 8. Ordering by day of the week



Observation. Monday (32 students) and Saturday (29) are the most frequently named ordering days, followed by Tuesday (25) and Friday (22). Wednesday (2) and Sunday (4), the days on which the mess serves non-veg meals, together account for 6 of 132 ordering-day mentions (4.5%), an average of 3.0 students per day against 25.2 on the other five days. This is consistent with students ordering less when the mess menu is more appealing, although the survey does not test the reason.

Figure 9. Social influence on ordering



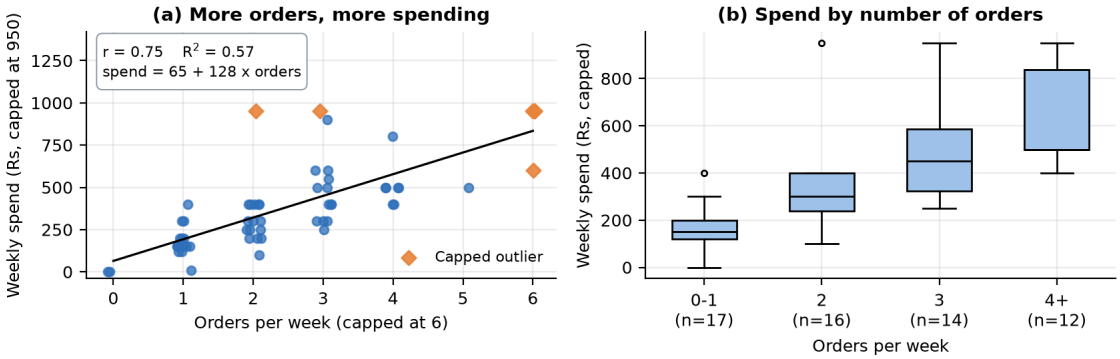
Observation. 44 (75%) students describe themselves as usually ordering with friends, 10 (17%) students as usually ordering alone, and 5 (8%) students as doing both about equally. Of the 159 orders placed by the group in a week, 106 (67%) are with friends. 40 students place more orders with friends than alone, 12 students place equal numbers, and 7 students place more alone; 28 (47%) students never order alone. Panel (b) compares the two counts within each self-described group, and the descriptions match the counts: students who say they order mostly with friends average 1.9 social orders against 0.6 solo orders a week, those who say mostly alone average 2.1 solo against 1.5 social, and the about-equally group averages 1.4 of each.

4.4 Relationships between mess habits and ordering habits

Strength of association is described with the rule of thumb: correlation of 0.6 or more is strong, 0.3 to 0.6 moderate, below 0.3 weak. Correlation shows that two things move together, not that one causes the other.

(a) Number of orders and spending

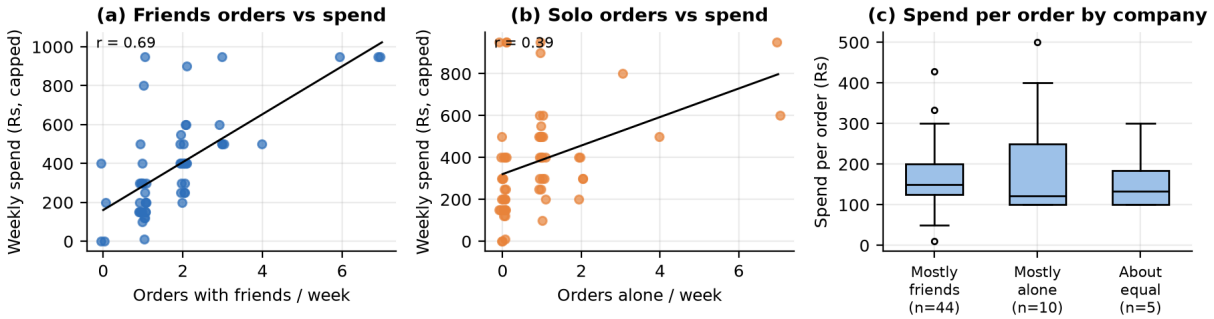
Figure 10. Number of orders and spending



Observation. More orders mean more spending. Median weekly spend is Rs 150 for 0-1 orders, Rs 300 for two, Rs 450 for three and Rs 500 for four or more (mean Rs 892, lifted by heavy spenders). The correlation is 0.75 (strong), with roughly Rs 128 extra spend per additional weekly order, in line with the typical Rs 150 per order.

(b) Social influence and spending

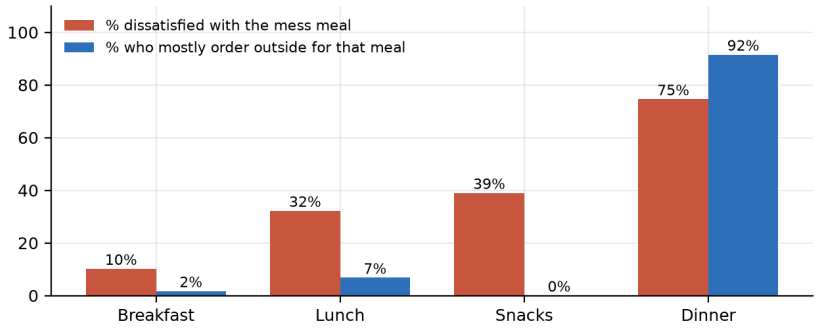
Figure 11. Social context and spending



Observation. Orders with friends are more closely tied to spending (correlation 0.69) than solo orders (0.39), which is expected because most orders are social. The company a student usually keeps does not follow the same pattern: students who order mostly alone actually order *more* often (3.6 a week) than those who order mostly with friends (2.5) or about equally (2.8), and they spend more per order (Rs 205 against Rs 161). The solo and equal groups are small (n = 10 and 5), so this is indicative only: social ordering accounts for most of the *volume* of orders, but it is not what makes an individual student order frequently.

(c) The least-liked meal is the meal students replace

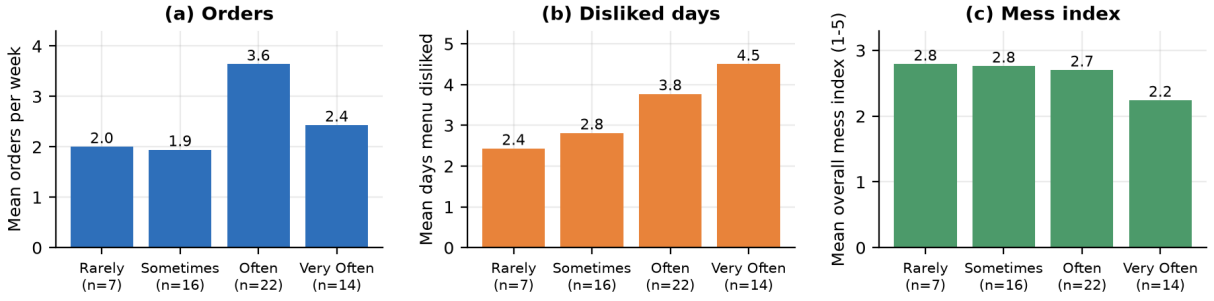
Figure 12. The least-liked meal is the one students replace



Observation. Dinner is where mess dissatisfaction (75%) and outside ordering (92%) meet. 40 of the 44 students who are dissatisfied with dinner mostly order outside for dinner. Breakfast, the best-rated meal, is almost never replaced.

(d) Menu influence

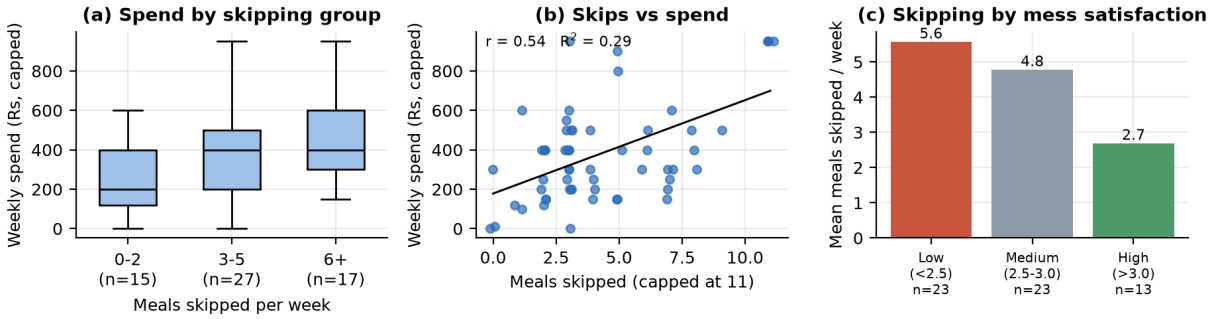
Figure 13. Menu influence on ordering (x-axis) vs orders, disliked days and mess rating



Observation. Mean orders per week rise with the stated influence of the menu: 2.0 for Rarely, 1.9 for Sometimes and 3.6 for Often. The rise is not uniform, as the Very Often group averages 2.4. The number of disliked days rises steadily across the same groups (2.4 to 4.5 days), and the Very Often group records the lowest overall mess rating (2.2 against about 2.8 for the other three groups).

(e) Meal skipping

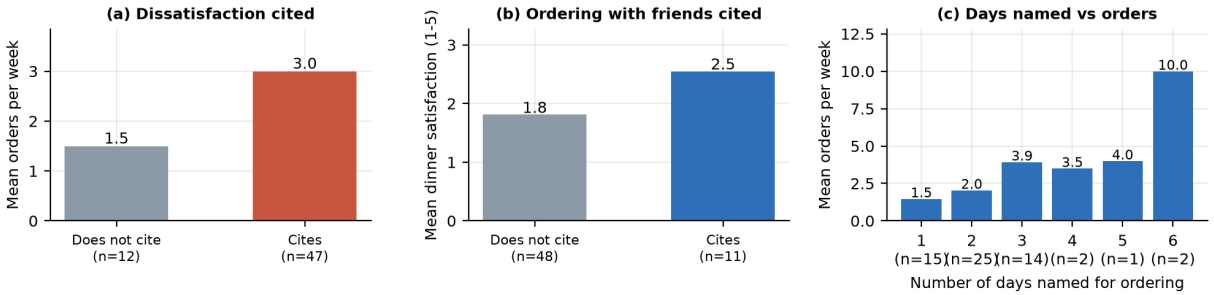
Figure 14. Meal skipping, spending and mess satisfaction



Observation. Weekly spending rises with the number of meals skipped. Median spend is Rs 200 for students skipping 0-2 meals, Rs 400 for 3-5 and Rs 400 for 6 or more (means Rs 240, Rs 389 and Rs 688); the correlation is 0.54, a moderate association. Skipping falls as mess satisfaction rises: students in the low-satisfaction group skip 5.6 meals a week against 2.7 in the high-satisfaction group. One result runs against expectation: students who usually order outside mess hours skip *fewer* meals (4.0) than those who do not (5.2), indicating that a skipped meal is not always replaced by an outside order.

(f) Stated reasons and behaviour

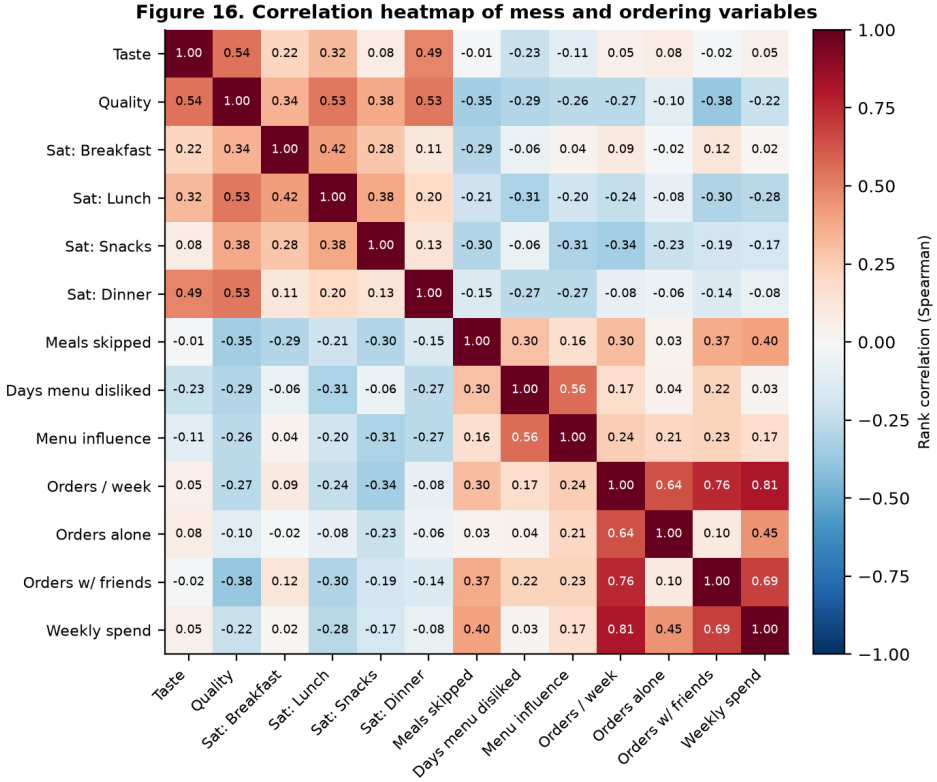
Figure 15. Stated reasons and ordering behaviour



Observation. Students who cite dissatisfaction as a reason order more often (3.0 orders a week) than those who do not (1.5). Students who cite ordering with friends rate mess dinner higher (2.5 against 1.8), which indicates that social ordering is driven by company rather than by dissatisfaction with the mess. Panel (c) serves as an internal consistency check: students who name more ordering days do place more orders per week, although the groups naming five or six days contain only 0 and 0 students, so the upper end of that trend rests on very few responses.

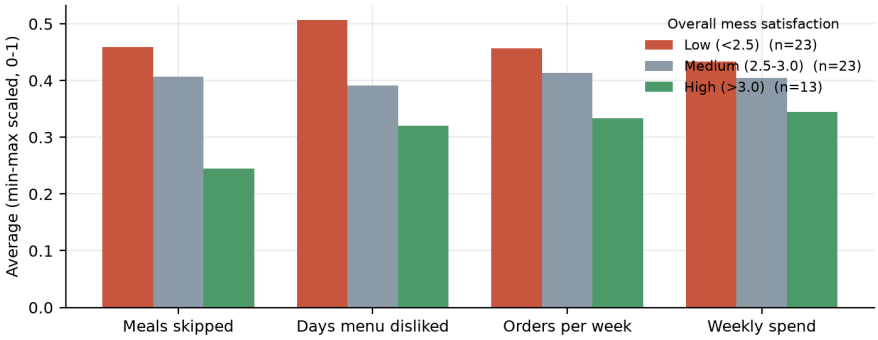
(g) The whole picture: correlation heatmap and behaviour profile

Figure 16. Correlation heatmap of mess and ordering variables



Strongest links. Orders per week and weekly spend (0.81); orders per week with orders with friends (0.76) and with orders alone (0.64); taste and quality (0.54); disliked days and menu influence (0.56); quality and dinner satisfaction (0.53). **Mess side:** better ratings of breakfast, snacks and dinner go with fewer skipped meals (about -0.29 to -0.15). **Weak links:** taste and orders per week (0.05) and dinner satisfaction and orders (-0.08) are close to zero. In other words, the mess ratings explain *whether* a student is unhappy and skips meals better than they explain *how many* orders the student places.

Figure 17. Behaviour profile by overall mess satisfaction



Observation. On the common 0-1 scale, students with low overall mess satisfaction skip more meals (0.46 vs 0.24), dislike the menu on more days (0.51 vs 0.32), order slightly more (0.46 vs 0.33) and spend slightly more (0.43 vs 0.34) than students with high satisfaction. The gap is large for habits inside the mess (skipping, disliked days) and small for ordering and spending.

4.5 Regression and a test of independence

Regression. Fitting weekly spend on the number of orders (both capped) gives **spend = 65 + 128 x orders**, with R^2 = 0.57. Each additional weekly order is associated with about Rs 128 more spending, close to the median of Rs 150 per order, and the number of orders alone explains 57% of the variation in what students spend. Meal skipping on its own explains much less (R^2 = 0.29).

Adding meal skipping and the overall mess index to the model raises R^2 to 0.63 (adjusted 0.61): spend = 16 + 109 x orders + 24 x meals skipped - 3 x mess index. Order count stays by far the strongest term; skipping adds a little, and the mess rating adds almost nothing once the other two are known. This supports the pattern seen in the heatmap: mess satisfaction is linked to *whether* a student skips meals rather than to *how much* they spend.

Chi-square test. We tested whether saying the daily menu often drives ordering is independent of being in the low mess-satisfaction group. Of the 36 students who answered Often or Very Often, 18 fall in the low group; of the 23 who answered Rarely or Sometimes, only 5 do. chi-square = 4.71 with 1 degree of freedom, above the 5% critical value of 3.84, so the two are not independent at the 5% level (Cramer's V = 0.28, a weak to moderate association). All expected counts exceed 5 (smallest 9.0), so the test is valid here.

5. Conclusion

5.1 Key findings

#	Finding	Evidence
1	Dinner is the weakest meal in the mess.	Dinner averages 1.95 out of 5 against 3.46 for breakfast. 75% of students are dissatisfied with dinner, against 10% for breakfast (Fig. 2).
2	Students replace the meal they like least.	92% order outside mainly for dinner, and 40 of the 44 students dissatisfied with dinner do so. Breakfast, the best-rated meal, is almost never replaced (Figs. 5, 12).
3	Spending is driven by how often a student orders, not by how much each order costs.	Median spend rises from Rs 150 to Rs 500 between the 0-1 and 4+ order groups. Order count alone explains 57% of the variation in spending, at about Rs 128 per extra order (Figs. 6, 10).
4	Ordering is mostly a group activity.	67% of all orders are placed with friends and 28 (47%) students never order alone. Group ordering drives the volume of orders, though students who order alone order slightly more often (Fig. 9).
5	Ordering almost stops on two days of the week.	Wednesday and Sunday account for 6 of 132 ordering-day mentions, an average of 3.0 students per day against 25.2 on the other five days (Fig. 8).
6	Dissatisfaction with the mess is the main reason given, and it is reflected in behaviour.	80% of students cite it, and they order 3.0 times a week against 1.5 for those who do not (Figs. 7, 15).
7	Skipping mess meals and disliking the menu go with more ordering and higher spending.	Median weekly spend is Rs 200 for students skipping 0-2 meals against Rs 400 for those skipping 6 or more. Students for whom the menu often drives ordering are also more likely to be in the low-satisfaction group (chi-square = 4.71, significant at 5%) (Figs. 13, 14).
8	Mess satisfaction explains skipping better than it explains ordering.	Satisfaction is moderately linked to meal skipping but only weakly to order counts, and adding the mess rating to the spending model changes little (R-squared 0.57 to 0.63) (Figs. 16, 17).

5.2 What this means

If mess management wants to reduce outside ordering, the data points first to **dinner**: improving its taste, quality and variety targets the meal that students are most unhappy with and most often replace. The low ordering on non-veg days suggests that menu appeal matters, and that a more appealing menu on other days could shift some of the ordering back to the mess. Because ordering is mostly done in groups, improvement may need to be visible to the whole friend circle to change habits.

5.3 Limitations

- The sample is small and non-random (59 students, convenience sampling, 85% third-year, a single mess), so the findings describe the respondents rather than the hostel population. Some subgroups are small (for example 10 students order mostly alone), and comparisons between them are indicative only.
- All measures are self-reported for a typical week rather than recorded, so the counts carry reporting error. Some responses required the consistency corrections described in Section 3.
- Extreme values were capped rather than removed. This limits their influence on the correlations but retains responses whose accuracy cannot be verified.
- The analysis is observational, so the associations reported do not establish cause. In particular, the survey does not test why ordering is low on Wednesdays and Sundays.

5.4 Possible extensions

A larger and more random sample, questions on the specific days and meals disliked, and the actual mess fees, would allow a direct comparison of what students pay the mess and what they pay outside.

Appendix: Questionnaire

The complete set of questions in the Google Form. All of them are used in the analysis.

#	Question	Answer format
1	How do you usually place your outside food orders?	Mostly alone / About equally alone and with friends / Mostly with friends
2	What is the main reason you order food from outside?	Multiple choice, select all: Dissatisfaction with mess food; Better taste; Craving for a particular food; Lack of variety in the mess; Ordering with friends; I skipped a mess meal; Convenience / too busy; Mess timing is inconvenient; Other
3	When do you most often order outside food?	Breakfast / Lunch / Dinner
4	On which day do you most often order outside food?	Multiple choice, select all: Monday to Sunday
5	Approximately how much do you spend on outside food orders in a typical week?	Number (Rs)
6	Do you usually order outside food outside the operating timings of the mess?	Yes / No
7	Do you usually order outside food because you skipped a mess meal?	Yes / No
8	In a typical week, how many times do you order outside food after skipping a mess meal?	Number
9	How satisfied are you with the taste of your mess food?	Rating 1 to 5
10	How often does the day's mess menu influence your decision to order outside food?	Rarely / Sometimes / Often / Very Often
11	How satisfied are you with each mess meal? (Breakfast, Lunch, Snacks, Dinner)	Very dissatisfied / Dissatisfied / Neutral / Satisfied / Very satisfied, for each meal
12	On average, how many mess meals do you skip in a typical week?	Number
13	Which mess meal do you prefer or like the most?	Breakfast / Lunch / Snacks / Dinner
14	How would you rate the overall quality and consistency of your mess food?	Rating 1 (very poor) to 5 (excellent / very consistent)
15	How many times do you order food from outside in a typical week?	Number
16	In a typical week, how many times do you order outside food alone?	Number
17	In a typical week, how many times do you order outside food with friends?	Number
18	In a typical week, on how many days do you dislike the mess menu or find it unappealing?	Number of days
19	What is your current year of study?	2nd year / 3rd year / 4th year / Other